DOC – GENIE

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Doc – Genie: Professional Python Function Docstring Generator

Introduction:

In modern software development, clear and well-structured documentation is essential for understanding and maintaining code. Documentation helps developers know how a program works, what inputs a function requires, and what outputs it produces. It also improves code readability and makes debugging and collaboration easier, especially when multiple developers work on the same project.

However, in many real-world projects, developers often skip writing proper documentation because creating docstrings manually for every function can be time-consuming and repetitive. As a result, many Python programs contain functions without clear explanations. This lack of documentation makes it difficult for other developers to understand the program logic, reuse the code, or maintain the software in the future.

To solve this problem, Doc-Genie is developed as an intelligent tool that automatically generates professional docstrings for Python functions. The system analyses Python code using AST (Abstract Syntax Tree) analysis to understand the internal structure of a function. Through this analysis, it identifies key elements such as the function name, parameters, return values, loops, and conditional statements.

Based on this information, Doc-Genie generates structured documentation in standard formats such as Google-style and NumPy-style docstrings, which are widely used in Python development.

The system also includes a user-friendly interface built using Gradio, allowing users to paste Python code or upload a .py file and instantly generate documentation.

Additionally, the generated documentation can be exported as PDF or TXT files, making it useful for project documentation and sharing with others.

Overall, Doc-Genie improves code readability, developer productivity, and documentation quality by automating the process of writing docstrings, making Python programs easier to understand and maintain.

Project Description:

Doc-Genie is a smart and intelligent tool designed to automatically generate professional documentation for Python functions. In modern software development, documentation is very important because it helps developers understand how a program works and how different functions should be used.

However, in many software projects, developers often skip writing proper documentation because it requires extra time and effort. Writing docstrings manually for every function can be repetitive and sometimes developers focus more on writing the code than documenting it. As a result, many Python programs contain functions without clear explanations.

Doc-Genie is developed to solve this problem by automatically generating clear and structured docstrings for Python functions. The system analyzes Python code using Abstract Syntax Tree (AST) analysis. AST is a powerful feature in Python that allows the system to understand the internal structure of the code. By analyzing the AST, the system can identify important elements such as the function name, parameters, default values, return values, loops, conditional statements, and operations used inside the function.

After analyzing the structure and behavior of the function, Doc-Genie automatically generates accurate and professional documentation. The generated documentation follows standard formats such as Google-style docstrings and NumPy-style docstrings, which are commonly used in Python development for writing clear and well-structured documentation.

The project also includes a user-friendly interface developed using Gradio. This interface allows users to easily interact with the system by pasting their Python code directly or uploading a .py file containing Python functions.

In addition to generating documentation, Doc-Genie also provides options to export the generated documentation as PDF or TXT files. These files can be used for software documentation, project reports, or sharing with other developers and team members.

Overall, Doc-Genie improves code readability, documentation quality, and developer productivity by automating the process of writing docstrings. It helps programmers save time, maintain better documentation, and make Python programs easier to understand, maintain, and share.

Objectives:

The main objectives of the **Doc-Genie** project are:

1. **To automatically generate professional docstrings** for Python functions in order to improve code documentation.
2. **To analyze Python code using Abstract Syntax Tree (AST)** so that the system can understand the structure and logic of functions.
3. **To reduce the time and effort required for manual documentation**, helping developers focus more on coding.
4. **To improve code readability and maintainability** by providing clear and structured explanations for functions.
5. **To support standard documentation formats** such as Google-style and NumPy-style docstrings.
6. **To provide a simple and user-friendly interface using Gradio**, where users can easily input Python code or upload .py files.
7. **To allow exporting generated documentation** in formats such as PDF or TXT for project reports and sharing.
8. **To assist students and developers** in understanding Python functions and improving software documentation practices.

Applications:

1. **Software Development Learning**

   Students learning Python programming can use Doc-Genie to understand how functions should be documented.
2. **Code Documentation**

   Developers can quickly generate documentation for Python projects.
3. **Educational Projects**

   Doc-Genie can be used in colleges and schools for programming assignments and projects.

4. **Team Collaboration**

   Proper documentation helps team members understand each other's code.

5. **Open-Source Development**

   Developers can use the tool to generate documentation for Python libraries.

6. **Code Maintenance**

   Well-documented code is easier to update, debug, and maintain.

System Architecture:

The system architecture of Doc-Genie explains how different components work together to generate documentation.

1. **User Interface (Frontend)**

   The frontend is created using Gradio, where users can:

   - Paste Python code
   - Upload .py files
   - Choose docstring style
   - View generated documentation

2. **Input Processing Module**

   This module reads and processes the python code entered by the user.

3. **AST Analysis Modules**

   The system uses Python AST to analyze the structure of the function and extract information such as parameters and return values.

4. **Docstring Generation Module**

   Based on the analysis, the system generates documentation in Google-style or NumPy-style format.

5. **Output Display Module**

   The generated documentation is displayed on the web interface.

6. **Export Module**

The system allows users to download documentation in PDF or TXT format.

Software Requirements:

- **Programming Language**
  - Python
- **Libraries**
  - Gradio
  - Transformers
  - Torch
  - Autopep8
  - Black
  - ReportLab
- **Development Environment**
  - Visual Studio Code (VS Code)
- **Operating System**
  - Windows
  - Linux
  - macOS
- **Web Browser**
  - Google Chrome
  - Microsoft Edge
  - Mozilla Firefox

Hardware Requirements:

- Processor: Intel Core i3 or higher
- RAM: Minimum 4 GB
- Storage: 10 GB free space
- Display: Monitor or laptop screen
- Input Devices: Keyboard and mouse
- Internet Connection for installing libraries

Project Workflow

- **Step 1:** User enters Python function or uploads .py file.
- **Step 2:** The system processes the input code.
- **Step 3:** AST analysis extracts function details.
- **Step 4:** Docstring generator creates structured documentation.
- **Step 5:** The generated docstring is displayed to the user.
- **Step 6:** Users can download documentation as PDF or TXT.

Features:

- Automatic docstring generation.
- AST-based code analysis'
- Google-style and NumPy-style documentation
- Simple user interface using Gradio
- Export documentation as PDF or TXT
- Beginner-friendly system
- Improves code readability

Future Enhancements:

- Support for more programming languages.
- Live preview of generated documentation.
- Integration with code editors like VS Code.
- Improved AI analysis for more accurate documentation.
- Cloud-based version for online use.

Conclusion:

Doc-Genie is a useful tool that helps developers automatically generate documentation for Python functions. By analyzing code using AST, the system can understand the function structure and produce accurate docstrings.

The tool improves code readability, saves time for developers, and ensures that Python programs remain well-documented. Doc-Genie demonstrates how automation and intelligent tools can improve the software development process.